THE Open GROUP

TOGAF® Standard – Architecture Deliverables Phase B

Version 1.0

October 2026

Barangay San Antonio Health Department Enterprise Architecture

Barangay San Antonio Health Center, Pasig City

Document Version History

Version Number	Version Date	Revised By	Description	Filename
1.0	October 22, 2026	UA&P Solution Architecture team	First issue, based on interviews (October 2–21, 2026) and verification answers (October 22, 2026)	B1 - Architecture Deliverables (Phase B).docx

Contents

1. Architectural Artifacts 5
2. Phase B: Business Architecture Artifacts 8
   - 2.1 Business Capability Map 8
   - 2.2 Organization/Actor Catalog 9
   - 2.3 Role Catalog 9
   - 2.4 Driver/Goal/Objective Catalog 10
   - 2.5 Actor/Role Matrix 10
   - 2.6 Business Service/Function Catalog 11
   - 2.7 Location Catalog 11
   - 2.8 Process Flow Diagrams 12
   - 2.9 Process/Event/Control/Product Catalog 14
     - 2.9.1 BP-2 Maintenance medicine pickup 14
     - 2.9.2 BP-4 Referral 15
     - 2.9.3 BP-6 Operation Timbang Plus 15
     - 2.9.4 BP-7 Dental visit 15
     - 2.9.5 BP-8 Budget and emergency purchase 16
     - 2.9.6 BP-9 Infectious waste disposal 16
   - 2.10 Business Event Catalog 16
   - 2.11 Business Gap Analysis 17

A Abbreviations 18

Tracking Information

Project Name	Barangay San Antonio Health Department Enterprise Architecture		
Prepared By	Rome Victor Mag-iba Sendin (Group Leader); Samuel Anthonius Badillo; Wilson Halim; Kevin Enrico Ang	**Document Version No.**	1.0
Title	Architecture Deliverables – Phase B: Business Architecture	**Document Version Date**	October 22, 2026
Reviewed By	Dr. Glynnis Lynn U. Bonilla, Medical Officer IV (Executive Sponsor)	**Review Date**	October 23, 2026

Distribution List

From	Date	Phone/Email
Rome Victor Mag-iba Sendin, Group Leader, UA&P Solution Architecture	October 22, 2026	romevictor.sendin@uap.asia / 09456315746

To	Action*	Due Date	Phone//Email
Dr. Glynnis Lynn U. Bonilla, Medical Officer IV	Approve	October 23, 2026	bsahealthcenter@gmail.com
Hon. Thomas Raymond U. Lising, Punong Barangay	Review	October 23, 2026	bsahealthcenter@gmail.com
Hon. Rachel Marie S. Rustia-Olarte, Co-Chair, Committee on Health & Nutrition	Review	October 23, 2026	bsahealthcenter@gmail.com
Mr. Joel Pira, Instructor, Solution Architecture (UA&P)	Review	October 23, 2026	joel.pira@uap.asia

* Action Types: Approve, Review, Inform, File, Action Required, Attend Meeting, Other (please specify)

1. Architectural Artifacts

While the Enterprise Metamodel is used to support the structuring of architectural information, most stakeholders do not need or wish to know the detail contained within the Architecture Content Framework in this manner. Therefore, the use of catalogs, matrices, and diagrams is introduced to facilitate the presentation of architectural information so that it may be used for reference and governance purposes more readily.

Catalogs are lists of building blocks of a specific type or related types, matrices are grids that show relationships between two or more entities, and diagrams are graphical renderings of architectural content.

In summary, the result of an Architecture Development Method (ADM) developed architecture consists of a number of defined Architecture Building Blocks populated into architecture catalogs, with relationships specified between those building blocks in architecture matrices, and/or also presented as diagrams, as appropriate to satisfy the stakeholder concerns.

The TOGAF® Standard provides a set of recommended artifacts, summarized as follows:

ADM Phase	**Artifact**
Preliminary Phase	Principles catalog
Phase A	Stakeholder Map matrix Value Chain diagram Solution Concept diagram Business Model diagram Business Capability map Value Stream map

ADM Phase	Artifact
Phase B	Organization/Actor catalog Driver/Goal/Objective catalog Role catalog Business Service/Function catalog Location catalog Process/Event/Control/Product catalog Contract/Measure catalog Business Capabilities catalog Value Stream catalog Value Stream Stages catalog Business Interaction matrix Actor/Role matrix Value Stream/Capability matrix Strategy/Capability matrix Capability/Organization matrix Business Footprint diagram Business Service/Information diagram Functional Decomposition diagram Product Lifecycle diagram Goal/Objective/Service diagram Business Use-Case diagram Organization Decomposition diagram Process Flow diagram Event diagram Business Capability map (updated) Value Stream map (updated) Organization map
Phase C: Data Architecture	Data Entity/Data Component catalog Data Entity/Business Function matrix Application/Data matrix Conceptual Data Diagram Logical Data Diagram Data Dissemination diagram Data Security diagram Data Migration diagram Data Lifecycle diagram
Phase C: Application Architecture	Application Portfolio catalog Interface catalog Application/Organization matrix Role/Application matrix Application/Function matrix Application Interaction matrix Application Communication diagram Application and User Location diagram Application Use-Case diagram Enterprise Manageability diagram Process/Application Realization diagram Software Engineering diagram Application Migration diagram Software Distribution diagram

ADM Phase	Artifact
Phase D	Technology Standards catalog Technology Portfolio catalog Application/Technology matrix Environments and Locations diagram Platform Decomposition diagram Processing diagram Networked Computing/Hardware diagram Network and Communications diagram
Phase E	Project Context diagram Benefits diagram
Requirements Management	Requirements catalog

2. Phase B: Business Architecture Artifacts

This section presents the Phase B artifacts committed in the Tailored Architecture Framework for the Barangay San Antonio Health Department, based on 16 interviews (October 2–21, 2026) and the health center's verification answers (October 22, 2026).

Tailoring note: the two Phase B templates (Architecture Requirements Specification and Architecture Roadmap) have no sections for business catalogs, matrices and diagrams. The Architecture Deliverables template, whose artifact table includes a Phase B row (Section 1), is therefore used to present them.

2.1 Business Capability Map

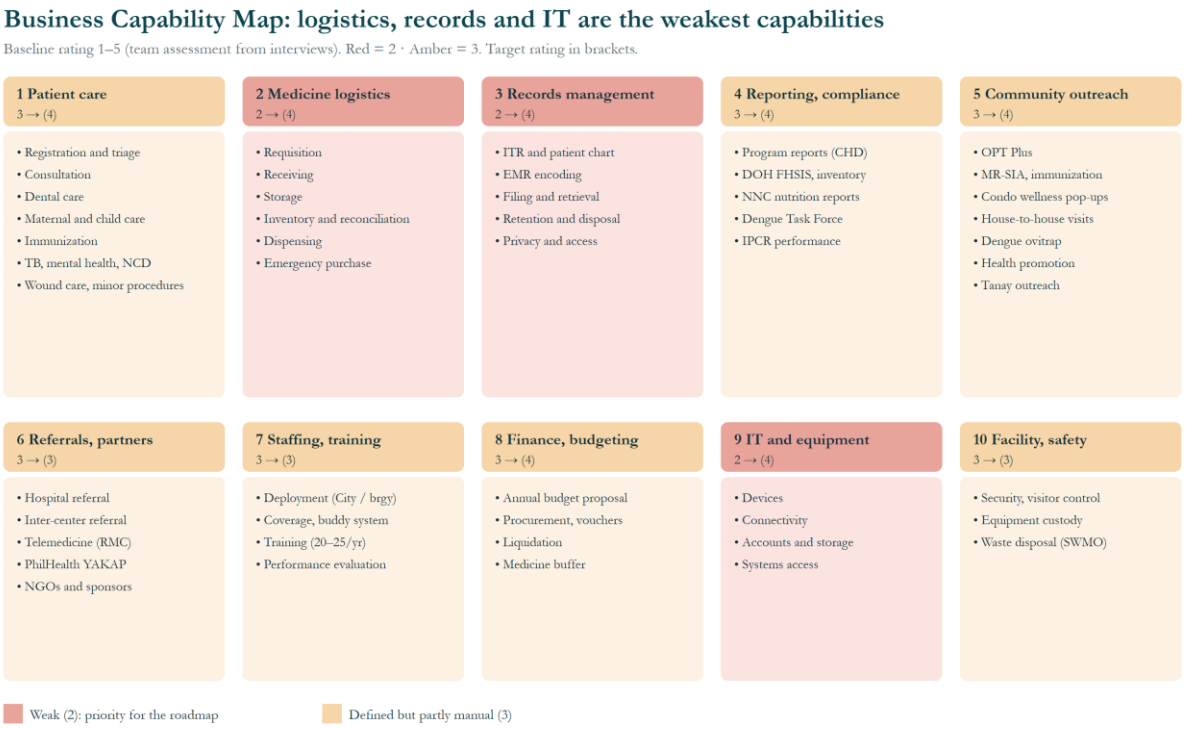


Figure 1. Business Capability Map (baseline → target ratings).

How to read it: each of the ten boxes is a level-1 capability; the bullets inside are the abilities it contains. The rating shows baseline → (target) on a 1–5 scale, assessed by the team from the interviews. Red = 2 (weak), amber = 3 (defined but partly manual).

- **Weakest (2 → 4):** medicine logistics (no stock alerts, partial deliveries), records management (~8,000 mixed charts, misfiling) and IT and equipment (10-year-old laptops, no frontline computers, personal accounts).
- **Working but manual (3 → 4):** patient care, reporting and compliance, community outreach, finance and budgeting.
- **No change planned (3 → 3):** referrals, staffing and training, facility and safety; these work adequately or are controlled by the City.

- **Takeaway:** the ratings set the roadmap priority: WP1 for logistics, WP4 for records, WP2, WP3 and WP7 for IT.

2.2 Organization/Actor Catalog

Organization unit	Type	Actors	Relationship to health center
Office of the City Mayor	External (City)	Mayor Victor Ma. Regis N. Sotto	Approves major system and policy changes
Pasig City Health Department	External (City)	City Health Officer Dr. Joseph R. Panaligan; CAO Dr. Philip Patrick C. Co; Records Officer Rosa C. Santos; Secretary Officer Algernon A. Victorino; program coordinators; city dental supervisor	Mandates, City staff, medicine supply, systems, report review
Barangay Council	Internal (Barangay)	Punong Barangay Thomas Raymond U. Lising; Kagawad Rachel Marie S. Rustia-Olarte (Co-Chair, Committee on Health & Nutrition)	Funding, approvals, programs
Barangay Treasurer's Office	Internal (Barangay)	Treasurer Rosalin S. Labayan; bookkeeper	Budget execution and liquidation
Barangay Nutrition Committee	Internal (Barangay)	Nutritionist (vice-chair), BNS, Kapitan, Kagawad	PPAN-aligned nutrition plans
Barangay IT team	Internal (Barangay)	IT staff (barangay hall)	Checks barangay equipment every 3 months
Health center	Core	16 on-site staff (see Role Catalog)	Care delivery
National agencies	External	DOH; National Nutrition Council; PhilHealth	Programs, standards, YAKAP, FHSIS
Care partners	External	Rizal Medical Center (incl. telemedicine); Pasig City General Hospital; Oranbo and Kapitolyo Health Centers; YAKAP pharmacy (Estancia)	Referrals and medicines
Community	External	Residents; ~63 condominiums; 5 Ortigas private preschools; Dengue Task Force; Tanay relocatees	Service users and access gatekeepers
LGU Solid Waste Management Office	External (City)	Collection team	Collects infectious waste twice a month

2.3 Role Catalog

Role	No.	Employer / status	Name(s)
Medical Officer IV (Rural Health Physician)	1	City; permanent	Dr. Glynnis Lynn U. Bonilla
Rural Health Dentist	1	City; permanent (Tue–Thu here; Mon/Fri at Oranbo)	Dr. Joseph P. Magpali, DMD
Rural Health Nurse	1	City; permanent	Ms. Nita N. Hardman, RN
Rural Health Midwife	1	City; permanent	Ms. Mafe B. Abong, RM
Nutritionist-Dietitian (Nutrition Officer II)	1	City; permanent (Tue–Thu here)	Mr. Christian Jay G. Vasquez, RND
Barangay Nutrition Scholar	1	City; permanent since about 2021	Ms. Mary Anne M. Verches
IT Officer / Medical Encoder	1	City; permanent	Mr. Tito F. Reyes

Role	No.	Employer / status	Name(s)
Pasig Health Aide	4	City; job order (paid and hired by CHD)	Ms. Janneth D. Correa (OIC), Ms. Rowena A. Intalan, Ms. Rosemarie T. Labandero, Ms. Procelyn L. Tatoy
Barangay Health Worker	3	Barangay; contractual (paid and hired by the barangay)	Ms. Joan Marie E. Barcenas, Mr. Perfecto S. Pestaño Jr., Ms. Josefina G. Roa (equipment custodian)
Security Officer	1	Barangay	Mr. Jeson C. Medez
Utility	1	Barangay	Ms. Vilma D. Lendio

2.4 Driver/Goal/Objective Catalog

Driver	Goal	Objective (proposed)
Prescriptions retyped in EMR and YAKAP (Dr. Bonilla, Dr. Magpali)	G1 Enter data once	Prescription entries 3 → 1
Three-month medicine shortage; ~₱150,000 barangay spend (verification)	G2 Keep medicines in stock	Stock-out months → 0
Misfiled charts; after-hours reporting (Ms. Abong, Nurse Hardman)	G3 Faster records and reports	Retrieval < 1 minute; no after-hours compilation
Work data on personal accounts (Mr. Vasquez, Nurse Hardman)	G4 Official, secure work tools	No work files on personal accounts
Queue complaints; condo resistance (Dr. Bonilla, Ms. Verches)	G5 Better patient flow and reach	Fewer complaints; higher condo coverage

2.5 Actor/Role Matrix

R = Responsible (does it) · A = Accountable · S = Supports · C = Consulted.

Activity	Doctor	Nurse	Midwife	Dentist	Nutri.	BNS	Encoder	Aides/ BHWs	Kap/ Kag	CHD
Registration and vitals		A	S					R		
Consultation and diagnosis	R/A	S	S							
Dental care				R/A						
EMR encoding	R		S	R			R			A
Dispensing		R/A						S		
Medicine requisition and inventory		R/A						S		S
Emergency medicine purchase	C	C							R/A	
Monthly program reports	A	R	R	R	R	S	S	S		C
OPT Plus			S		A	R		S	S	
MR-SIA / immunization		R/A	R					S		C
Condo wellness pop-up	R	S	S					S	A	
Dengue ovitrap		A						R		C
Referral and escort	A	R						S		
Annual health budget	C				C				R/A	
Process or system changes	R	C	C	C			C		C	A

Activity	Doctor	Nurse	Midwife	Dentist	Nutri.	BNS	Encoder	Aides/ BHWs	Kap/ Kag	CHD
Equipment custody		A						R (Roa)		

2.6 Business Service/Function Catalog

Service	Schedule / volume	Function owner
Medical consultation	Mon/Tue/Fri 8–11 AM; 360 last month	Dr. Bonilla
Maintenance-medicine dispensing	Mon/Wed/Fri 1–4 PM; 297 pickups last month	Nurse Hardman
Immunization	Wednesdays; MR-SIA campaigns	Nurse Hardman; Ms. Abong
Prenatal and family planning	Consultation days; house-to-house visits	Ms. Abong
Dental	Tue–Thu, scheduled; 10–15 patients a day	Dr. Magpali
Nutrition (OPT Plus, counseling, feeding)	OPT Jan–Mar; 798 children 0–59 months	Mr. Vasquez; Ms. Verches
TB, mental health, smoking cessation, NCD	Ongoing program care	Nurse Hardman
Referral	About 30 a month	Dr. Bonilla; Nurse Hardman
Dengue ovitrap	Report every 2 weeks	Aides and BHWs
PhilHealth YAKAP registration and prescriptions	Per patient	Dr. Bonilla; Mr. Reyes

2.7 Location Catalog

Location	Use
Health center (Barangay San Antonio)	All clinic services, records room, medicine storage
Barangay hall	Barangay IT team; Treasurer; Council
Old Pasig City Hall	Hard-copy report submission
Condominiums (~63), 5 Ortigas private preschools, daycare, streets	Outreach, OPT Plus, MR-SIA, wellness pop-ups
Tanay, Rizal relocation site	Monthly OPT Plus visits, January–March
Rizal Medical Center, PCGH, Oranbo and Kapitolyo Health Centers	Referral destinations

2.8 Process Flow Diagrams

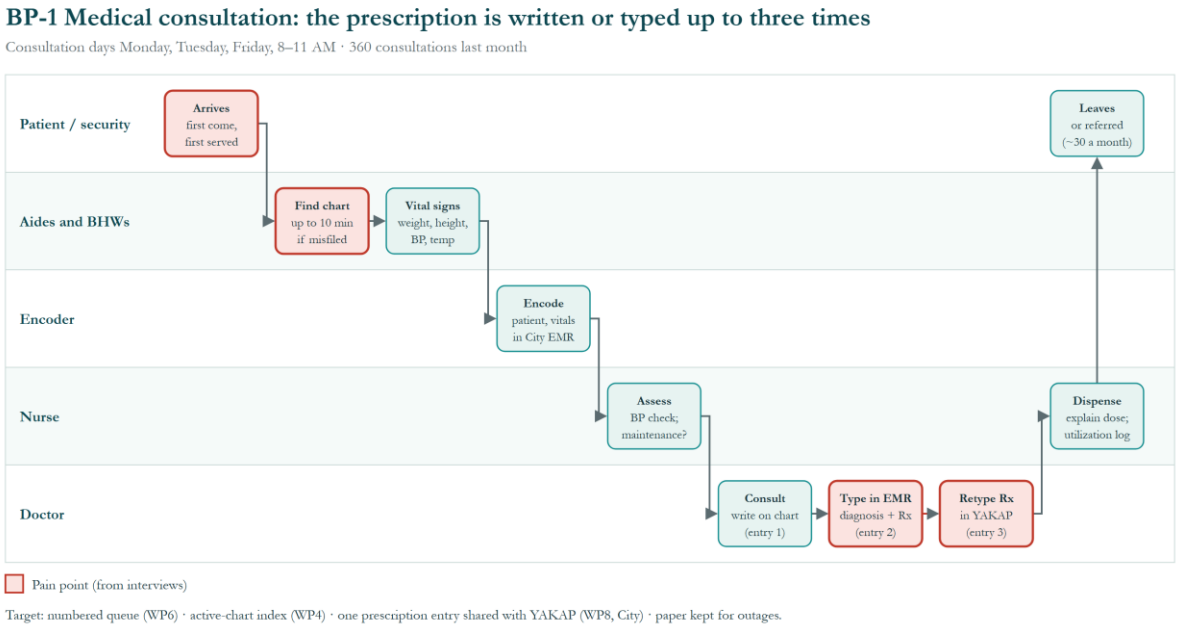


Figure 2. BP-1 Medical consultation.

How to read it: each row ("swimlane") is one actor: patient/security, aides and BHWs, encoder, nurse and doctor. Boxes run left to right in time order; red boxes are pain points. Volume: 360 consultations last month (Monday, Tuesday and Friday, 8–11 AM).

1. The patient arrives and is served first come, first served.
2. An aide or BHW finds the chart (up to 10 minutes if misfiled) and takes vital signs.
3. The encoder enters the patient and vitals in the City EMR.
4. The nurse assesses (BP check; maintenance patient?).
5. The doctor consults and writes on the chart (entry 1), types the diagnosis and prescription in the EMR (entry 2), and retypes the prescription in PhilHealth YAKAP (entry 3).
6. The nurse dispenses and explains the dose, recording it in the utilization log; the patient leaves or is referred (about 30 referrals a month).

Takeaway: retyping a PhilHealth patient's 5–10 medicines is avoidable work. Targets: numbered queue (WP6), active-chart index (WP4), one shared prescription entry (WP8, City); paper stays as the outage fallback.

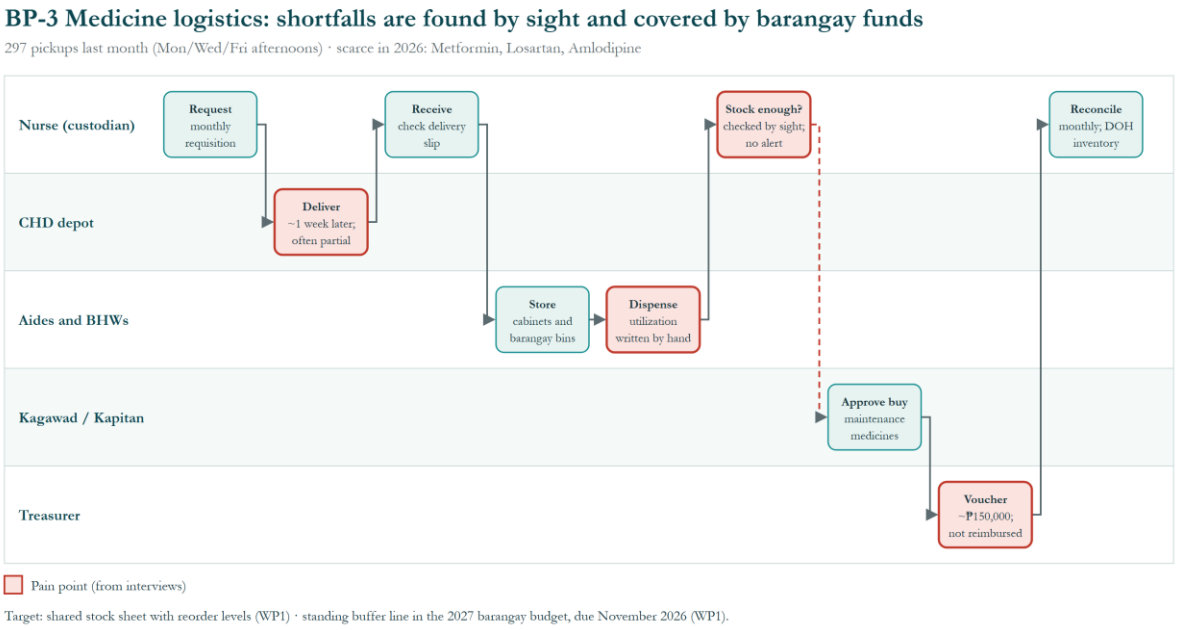


Figure 3. BP-3 Medicine logistics.

How to read it: rows are the nurse (medicine custodian), the City Health Department depot, aides and BHWs, the Kagawad/Kapitan and the Treasurer. Volume: 297 maintenance-medicine pickups last month.

7. The nurse sends a monthly requisition to the CHD.
8. The CHD delivers about a week later, often incomplete; the nurse checks the delivery slip and stores stock in cabinets and barangay bins.
9. Aides and BHWs dispense and write each issue in the utilization log by hand.
10. Whether stock is enough is checked by sight; there is no alert.
11. When short, the Kagawad and Kapitan approve a barangay purchase and the Treasurer issues a voucher (about ₱150,000 in 2026, not reimbursed).
12. The nurse reconciles stock monthly for the DOH inventory report.

Takeaway: shortages are found late and paid for unplanned. Targets: a shared stock sheet with reorder levels and a standing buffer line in the 2027 barangay budget (both WP1).

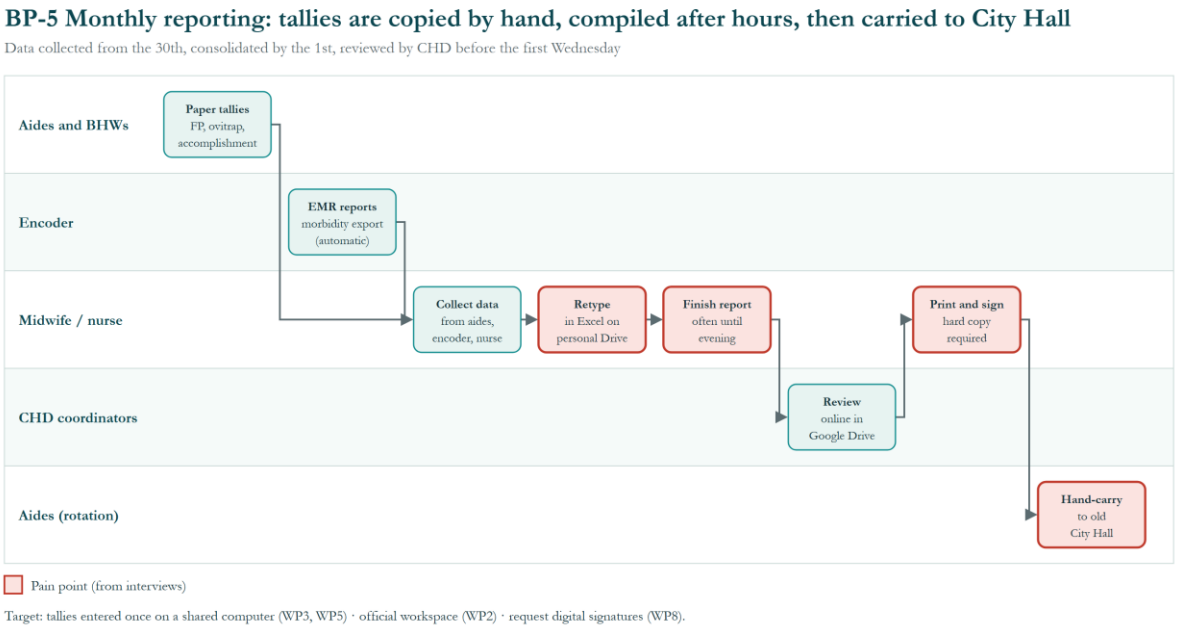


Figure 4. BP-5 Monthly reporting.

How to read it: rows are aides and BHWs, the encoder, the midwife/nurse, CHD coordinators and aides on rotation. Timing: data collected from the 30th, consolidated by the 1st, reviewed by the CHD before the first Wednesday.

13. Aides and BHWs keep paper tallies (family planning, ovitrap, accomplishment); the EMR produces the morbidity export automatically.
14. The midwife or nurse collects data from aides, the encoder and the nurse.
15. Figures are retyped into Excel on personal Google Drive and finished, often into the evening.
16. CHD coordinators review the report online in Google Drive.
17. A hard copy is printed and signed (required) and carried by aides on rotation to the old City Hall.

Takeaway: only the EMR export is automatic; reports are never late, but at the cost of after-hours work. Targets: shared computers and a single tally sheet (WP3, WP5), official workspace (WP2), and a request to accept digitally signed reports (WP8).

2.9 Process/Event/Control/Product Catalog

2.9.1 BP-2 Maintenance medicine pickup

Trigger: Monday, Wednesday or Friday, 1–4 PM.

#	Actor	Activity	Tool / record
1	Aide/BHW	Checks patient booklet and enrollment	Booklet, ITR
2	Aide/BHW	Takes BP; records in monthly monitoring	Maintenance monitoring sheet
3	Aide/BHW, nurse	Dispenses one month's supply (10–20 pieces during shortages)	Utilization log

#	Actor	Activity	Tool / record
4	Patient	Signs utilization record	Utilization log
5	Encoder	Encodes dispensing	City EMR

Pain points: 297 pickups last month; supply often short; quantity limits.

Target change: Digital stock view with reorder levels; buffer stock.

2.9.2 BP-4 Referral

Trigger: Patient needs a laboratory, hospital care, or a doctor when the physician is absent.

#	Actor	Activity	Tool / record
1	Doctor or nurse	Decides on referral; writes referral	Referral slip
2	Aide/BHW	Brings patient record; escorts by barangay vehicle when needed	Patient chart
3	Receiving facility	Rizal Medical Center, PCGH, Oranbo or Kapitolyo, or telemedicine	—

Pain points: About 30 a month; no referral log or outcome feedback.

Target change: Referral register with outcome follow-up.

2.9.3 BP-6 Operation Timbang Plus

Trigger: January–March; preparation year-round.

#	Actor	Activity	Tool / record
1	Nutritionist	Plans schedule; letters signed by Dr. Bonilla, Kapitan and Dr. Panaligan	Letters
2	BNS + aides/BHWs (3–5 per team)	Weigh and measure 0–59 months; vitamin A, deworming	e-OPT form, master list
3	BNS	Visits Tanay monthly Jan–Mar; follow-ups for missed children	Master list
4	BNS	Encodes results	NNC e-OPT tool, Google Sheets
5	Nutritionist	Validates (at least 80% coverage); reports to city → region → national	Google Drive

Pain points: 798 children; condo access refused; Tanay distance; 2–3 GB files on 15 GB personal storage.

Target change: Field laptop for direct entry; official storage.

2.9.4 BP-7 Dental visit

Trigger: Scheduled appointment (walk-in for new patients).

#	Actor	Activity	Tool / record
1	Patient	Books a 15–60 minute slot	Appointment list

#	Actor	Activity	Tool / record
2	Dental assistant	Pulls record; takes BP and temperature	Dental record
3	Dentist	Treats (preventive, ART, extraction)	Dental record
4	Dentist / assistant	Encodes dental chart; files PhilHealth claim	Dental eMED + PhilHealth

Pain points: Separate dental system; low preventive uptake.

Target change: Online/QR booking (dentist's plan); integration request to CHD.

2.9.5 BP-8 Budget and emergency purchase

Trigger: Annual budget cycle (proposals due November 2026) or an urgent need.

#	Actor	Activity	Tool / record
1	Health center	Requests via Kagawad Rustia or Kapitan	Request
2	Barangay Council	Prepares budget; Kapitan approves; City Council adopts	Budget
3	Treasurer	Posting, bidding, voucher, check, liquidation	Ledger; Landbank
4	Nutritionist	Submits 3-year nutrition plan (2026: ₱450,301; 2027: ₱650,809)	PPAs

Pain points: Speed depends on amount; no standing medicine buffer.

Target change: Costed requests in the 2027 budget; buffer line.

2.9.6 BP-9 Infectious waste disposal

Trigger: Waste generated during care.

#	Actor	Activity	Tool / record
1	Utility / staff	Segregate and store infectious and hazardous waste	—
2	LGU Solid Waste Management Office	Collects twice a month	—

Pain points: None reported.

Target change: No change; documented as part of facility and safety.

2.10 Business Event Catalog

Event	When	Impact
Consultation days	Mon/Tue/Fri 8–11 AM	Peak workload
Immunization	Wednesdays	Nurse and midwife busy
Maintenance-medicine pickup	Mon/Wed/Fri 1–4 PM	Dispensing and encoding load
Ovitrap report	Every 2 weeks	Field and reporting
Family planning unmet-need report	3rd week of the month	Reporting
Monthly reports	30th → 1st; CHD review before the first Wednesday	After-hours compilation

Event	When	Impact
OPT Plus / MR-SIA	January–March	Whole team in the field
Infectious waste pickup	Twice a month (LGU SWMO)	Waste stored until pickup
Barangay budget proposals	Due November 2026	Funding window
BNS convention	October/November (~₱2,000, paid by the Kapitan)	BNS away
City equipment audit	About every 6 months	City-issued devices checked

2.11 Business Gap Analysis

ID	Baseline	Target	Gap	Controlled by
BG-1	Prescription written on the chart, typed in the EMR, then retyped in PhilHealth YAKAP	Entered once	No EMR–YAKAP link	City / vendor
BG-2	Medicine stock tracked by hand and checked by sight	Continuous stock view with reorder alerts	No tracking tool	Health center
BG-3	Barangay buys medicine shortfalls ad hoc (about ₱150,000 in 2026, not reimbursed)	Planned buffer stock	No standing budget line	Barangay
BG-4	About 8,000 charts, active and inactive mixed; misfiled charts take up to 10 minutes	Active-chart index; inactive archived	No filing standard or records clerk	Health center
BG-5	Reports compiled after hours; signed hard copies hand-carried to City Hall	Single tally source; digital submission	Redundant forms; hard-copy rule	Health center / City
BG-6	First come, first served; complaints about order	Numbered queue	No queue process	Health center (CHO approval)
BG-7	Aides and BHWs record on paper only (tech literacy about 2/5)	Shared computers and basic training	No devices; low skills	Barangay
BG-8	Condo residents resist government services; letters often refused	Higher reach via partnerships	Access and trust	Barangay
BG-9	Nurse and doctor carry heavy paperwork; one encoder for all systems	Encoding support	Encoder capacity	City / Barangay

A Abbreviations

Abbreviations, acronyms and reference codes used in this document and its figures, in alphabetical order.

Abbreviation	Meaning
ADM	Architecture Development Method, the step-by-step method of the TOGAF Standard
ART	Atraumatic Restorative Treatment (a dental filling technique)
BG-1 …	Business gap reference number (Phase B)
BHW	Barangay Health Worker
BNS	Barangay Nutrition Scholar
BP	Blood pressure
BP-1 …	Business process reference number
Brgy	Barangay
CAO	Chief Administrative Officer
CHD	City Health Department (Pasig City)
CHO	City Health Officer
DMD	Doctor of Dental Medicine
DOH	Department of Health
e-OPT	Electronic Operation Timbang Plus tool of the National Nutrition Council
eMed / eMED	Name of the City's electronic medical record system; “dental eMED” is its separate dental system
EMR	Electronic Medical Record
FHSIS	Field Health Services Information System (DOH)
FP	Family planning
G1 … G5	Goal reference number
GB	Gigabyte (a unit of file size)
Hon.	Honorable (title of elected officials)
IPCR	Individual Performance Commitment and Review
IT	Information technology
ITR	Individual Treatment Record (the patient's registration form)
Kap / Kag	Kapitan (Punong Barangay) / Kagawad (barangay council member)
LGU	Local Government Unit
MR-SIA	Measles-Rubella Supplemental Immunization Activity
NCD	Non-communicable disease (e.g., hypertension, diabetes)
NGO	Non-government organization
NNC	National Nutrition Council
Nutri.	Nutritionist-Dietitian (table heading)
OIC	Officer-in-Charge
OPT / OPT Plus	Operation Timbang Plus: yearly weighing and measuring of children aged 0–59 months
PCGH	Pasig City General Hospital
PhilHealth	Philippine Health Insurance Corporation
PPAN	Philippine Plan of Action for Nutrition
PPAs	Programs, projects and activities
QR	Quick Response (a scannable square barcode)
RM	Registered Midwife
RMC	Rizal Medical Center
RN	Registered Nurse
RND	Registered Nutritionist-Dietitian

Abbreviation	Meaning
Rx	Prescription
SWMO	Solid Waste Management Office (City of Pasig)
TB	Tuberculosis
TOGAF	The Open Group Architecture Framework (the TOGAF® Standard)
UA&P	University of Asia and the Pacific
WP1 … WP9	Work package (a group of related changes delivered together)
YAKAP	PhilHealth Yaman ng Kalusugan Program (primary-care benefit)